

Barrett Viator - Capstone EDA API Work

August 29, 2025

```
[ ]: # The code in this file was produced by Barrett Viator
```

```
[1]: import pandas as pd
import wbgapi as wb
import requests
import openpyxl
import pycountry
```

```
[37]: #Series Codes of Selected Topics from World Bank - BJV
```

```
wb_series_codes = [
    "economy",
    "time",
    "NY.GDP.PCAP.KD.ZG",
    "NY.GNP.PCAP.KD.ZG",
    "SE.XPD.TOTL.GD.ZS",
    "SE.ADT.LITR.ZS",
    "SE.COM.DURS",
    "SE.LPV.PRIM",
    "SE.LPV.PRIM.SD",
    "SE.SEC.CUAT.LO.ZS",
    "SE.SEC.CUAT.PO.ZS",
    "SE.SEC.CUAT.UP.ZS",
    "SE.TER.CUAT.BA.ZS",
    "SE.TER.CUAT.DO.ZS",
    "SE.TER.CUAT.MS.ZS",
    "SE.TER.CUAT.ST.ZS",
    "AG.PRD.FOOD.XD",
    "EN.POP.DNST",
    "EN.POP.SLUM.UR.ZS",
    "SP.RUR.TOTL.ZG",
    "EG.ELC.ACCS.ZS",
    "ER.H2O.FWST.ZS",
    "FX.OWN.TOTL.ZS",
    "SN.ITK.MSFI.ZS",
    "SH.XPD.CHEX.PC.CD",
    "SH.XPD.GHED.PC.CD",
    "SH.STA.WASH.P5",
```

```

"SP.DYN.LE00.IN",
"SH.UHC.SRVS.CV.XD",
"IT.NET.BBND.P2",
"IT.NET.USER.ZS",
"GB.XPD.RSDV.GD.ZS",
"IS.SHP.GCNW.XQ",
"SI.POV.GINI",
"VC.IHR.PSRC.P5",
"CC.EST",
"GE.EST",
"PV.EST",
"RL.EST",
"VA.EST",
"SM.POP.TOTL.ZS",
"SI.POV.GINI"
]

print("Collecting information provided by World Bank Series Codes:\n")

```

Collecting information provided by World Bank Series Codes:

```

[38]: #create a data frame from the World Bank API - BJV
wb_df = wb.data.DataFrame(
    wb_series_codes,
    economy='all',
    time=range(1995,2024),
    columns='series'
)

print("Data retrieval complete: Displaying first 5 rows:")
print(wb_df.head())

print("\nDataFrame Info:")
wb_df.info()

print(f"\nDataFrame shape: {wb_df.shape} (rows, columns)")

```

Data retrieval complete: Displaying first 5 rows:

economy	time	AG.PR.D.FOOD.XD	CC.EST	EG.ELC.ACCS.ZS	EN.POP.DNST	\
ABW	YR1995	NaN	NaN	100.0	443.361111	
	YR1996	NaN	NaN	100.0	461.227778	
	YR1997	NaN	NaN	100.0	479.450000	
	YR1998	NaN	NaN	100.0	491.394444	
	YR1999	NaN	NaN	100.0	498.105556	

		EN.POP.SLUM.UR.ZS	ER.H2O.FWST.ZS	FX.OWN.TOTL.ZS	\
economy	time				
ABW	YR1995	NaN	NaN	NaN	
	YR1996	NaN	NaN	NaN	
	YR1997	NaN	NaN	NaN	
	YR1998	NaN	NaN	NaN	
	YR1999	NaN	NaN	NaN	

		GB.XPD.RSDV.GD.ZS	GE.EST	IS.SHP.GCNW.XQ	...	\
economy	time				...	
ABW	YR1995	NaN	NaN	NaN	...	
	YR1996	NaN	NaN	NaN	...	
	YR1997	NaN	NaN	NaN	...	
	YR1998	NaN	NaN	NaN	...	
	YR1999	NaN	NaN	NaN	...	

		SH.UHC.SRVS.CV.XD	SH.XPD.CHEX.PC.CD	SH.XPD.GHED.PC.CD	\
economy	time				
ABW	YR1995	NaN	NaN	NaN	
	YR1996	NaN	NaN	NaN	
	YR1997	NaN	NaN	NaN	
	YR1998	NaN	NaN	NaN	
	YR1999	NaN	NaN	NaN	

		SI.POV.GINI	SM.POP.TOTL.ZS	SN.ITK.MSFI.ZS	SP.DYN.LE00.IN	\
economy	time					
ABW	YR1995	NaN	27.729502	NaN	72.832	
	YR1996	NaN	NaN	NaN	72.856	
	YR1997	NaN	NaN	NaN	72.904	
	YR1998	NaN	NaN	NaN	72.940	
	YR1999	NaN	NaN	NaN	72.857	

		SP.RUR.TOTL.ZG	VA.EST	VC.IHR.PSRC.P5
economy	time			
ABW	YR1995	3.610860	NaN	7.787151
	YR1996	4.750120	NaN	NaN
	YR1997	4.666101	NaN	NaN
	YR1998	3.245730	NaN	NaN
	YR1999	2.132050	NaN	2.302370

[5 rows x 39 columns]

DataFrame Info:

```
<class 'pandas.core.frame.DataFrame'>
```

MultiIndex: 7714 entries, ('ABW', 'YR1995') to ('ZWE', 'YR2023')

Data columns (total 39 columns):

#	Column	Non-Null Count	Dtype
---	--------	----------------	-------

```

---  -----  -----  -----
0  AG.PRD.FOOD.XD      6642 non-null   float64
1  CC.EST              4988 non-null   float64
2  EG.ELC.ACCS.ZS      7195 non-null   float64
3  EN.POP.DNST         7272 non-null   float64
4  EN.POP.SLUM.UR.ZS   2435 non-null   float64
5  ER.H2O.FWST.ZS      4496 non-null   float64
6  FX.OWN.TOTL.ZS      653 non-null    float64
7  GB.XPD.RSDV.GD.ZS   3062 non-null   float64
8  GE.EST              4964 non-null   float64
9  IS.SHP.GCNW.XQ      2628 non-null   float64
10 IT.NET.BBND.P2      5104 non-null   float64
11 IT.NET.USER.ZS      5671 non-null   float64
12 NY.GDP.PCAP.KD.ZG   7337 non-null   float64
13 NY.GNP.PCAP.KD.ZG   4628 non-null   float64
14 PV.EST              5025 non-null   float64
15 RL.EST              5078 non-null   float64
16 SE.ADT.LITR.ZS      2060 non-null   float64
17 SE.COM.DURS         6027 non-null   float64
18 SE.LPV.PRIM         364 non-null    float64
19 SE.LPV.PRIM.SD      364 non-null    float64
20 SE.SEC.CUAT.LO.ZS   1926 non-null   float64
21 SE.SEC.CUAT.PO.ZS   1406 non-null   float64
22 SE.SEC.CUAT.UP.ZS   1996 non-null   float64
23 SE.TER.CUAT.BA.ZS   1314 non-null   float64
24 SE.TER.CUAT.DO.ZS   1150 non-null   float64
25 SE.TER.CUAT.MS.ZS   1124 non-null   float64
26 SE.TER.CUAT.ST.ZS   1712 non-null   float64
27 SE.XPD.TOTL.GD.ZS   4861 non-null   float64
28 SH.STA.WASH.P5      231 non-null    float64
29 SH.UHC.SRVS.CV.XD   1428 non-null   float64
30 SH.XPD.CHEX.PC.CD   5461 non-null   float64
31 SH.XPD.GHED.PC.CD   5448 non-null   float64
32 SI.POV.GINI         2006 non-null   float64
33 SM.POP.TOTL.ZS      1300 non-null   float64
34 SN.ITK.MSFI.ZS      1021 non-null   float64
35 SP.DYN.LE00.IN       7685 non-null   float64
36 SP.RUR.TOTL.ZG      7345 non-null   float64
37 VA.EST              5051 non-null   float64
38 VC.IHR.PSRC.P5      3735 non-null   float64

```

dtypes: float64(39)

memory usage: 2.3+ MB

DataFrame shape: (7714, 39) (rows, columns)

```
[39]: # Index needs to be reset in order for 'economy' and 'time' to be accessed and
      ↪renamed - BJV
```

```

wb_df = wb_df.reset_index()

wb_df = wb_df.rename(columns={'economy': 'ISO3', 'time': 'year'})

# Remove 'YR' from the 'year' column to leave just four digit years - BJV
wb_df['year'] = wb_df['year'].astype(str).str.replace('YR', '', regex=False)

# convert the year to a numeric from string for errors arising later - BJV
wb_df['year'] = pd.to_numeric(wb_df['year'], errors='coerce').astype('Int64')

print("\nDataFrame Info:")
wb_df.info()

```

DataFrame Info:

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 7714 entries, 0 to 7713

Data columns (total 41 columns):

#	Column	Non-Null Count	Dtype
0	ISO3	7714 non-null	object
1	year	7714 non-null	Int64
2	AG.PR.D.FOOD.XD	6642 non-null	float64
3	CC.EST	4988 non-null	float64
4	EG.ELC.ACCS.ZS	7195 non-null	float64
5	EN.POP.DNST	7272 non-null	float64
6	EN.POP.SLUM.UR.ZS	2435 non-null	float64
7	ER.H2O.FWST.ZS	4496 non-null	float64
8	FX.OWN.TOTL.ZS	653 non-null	float64
9	GB.XPD.RSDV.GD.ZS	3062 non-null	float64
10	GE.EST	4964 non-null	float64
11	IS.SHP.GCNW.XQ	2628 non-null	float64
12	IT.NET.BBND.P2	5104 non-null	float64
13	IT.NET.USER.ZS	5671 non-null	float64
14	NY.GDP.PCAP.KD.ZG	7337 non-null	float64
15	NY.GNP.PCAP.KD.ZG	4628 non-null	float64
16	PV.EST	5025 non-null	float64
17	RL.EST	5078 non-null	float64
18	SE.ADT.LITR.ZS	2060 non-null	float64
19	SE.COM.DURS	6027 non-null	float64
20	SE.LPV.PRIM	364 non-null	float64
21	SE.LPV.PRIM.SD	364 non-null	float64
22	SE.SEC.CUAT.LO.ZS	1926 non-null	float64
23	SE.SEC.CUAT.PO.ZS	1406 non-null	float64
24	SE.SEC.CUAT.UP.ZS	1996 non-null	float64
25	SE.TER.CUAT.BA.ZS	1314 non-null	float64
26	SE.TER.CUAT.DO.ZS	1150 non-null	float64

```

27 SE.TER.CUAT.MS.ZS 1124 non-null float64
28 SE.TER.CUAT.ST.ZS 1712 non-null float64
29 SE.XPD.TOTL.GD.ZS 4861 non-null float64
30 SH.STA.WASH.P5 231 non-null float64
31 SH.UHC.SRVS.CV.XD 1428 non-null float64
32 SH.XPD.CHEX.PC.CD 5461 non-null float64
33 SH.XPD.GHED.PC.CD 5448 non-null float64
34 SI.POV.GINI 2006 non-null float64
35 SM.POP.TOTL.ZS 1300 non-null float64
36 SN.ITK.MSFI.ZS 1021 non-null float64
37 SP.DYN.LE00.IN 7685 non-null float64
38 SP.RUR.TOTL.ZG 7345 non-null float64
39 VA.EST 5051 non-null float64
40 VC.IHR.PSRC.P5 3735 non-null float64
dtypes: Int64(1), float64(39), object(1)
memory usage: 2.4+ MB

```

```

[40]: #save this data frame as an xlsx - BJV
wb_df.to_excel("wb_api_dataframe_full3.xlsx", index=True)

```

```

[41]: # The API call would throw an error if I tried to do more than
#two years at a time, so individual df were made and concat to resolve this -BJV

```

```

[42]: # UNDP API Key and Request - BJV

undp_url = "https://hdrdata.org/api/CompositeIndices/query?
↳apikey=HDR-K0gLmhxaLjacLvGRi4oTPARBLx54ubNe&year=2022,2023"

response = requests.get(undp_url)

if response.status_code == 200:
    data = response.json()

df1 = pd.DataFrame(data)

```

```

[43]: # UNDP API Key and Request - BJV

undp_url2 = "https://hdrdata.org/api/CompositeIndices/query?
↳apikey=HDR-K0gLmhxaLjacLvGRi4oTPARBLx54ubNe&year=2020,2021"

response2 = requests.get(undp_url2)

if response2.status_code == 200:
    data2 = response2.json()

df2 = pd.DataFrame(data2)

```

[44]: *# UNDP API Key and Request - BJV*

```
undp_url3 = "https://hdrdata.org/api/CompositeIndices/query?
↳apikey=HDR-K0gLmhxaLjacLvGRi4oTPARBLx54ubNe&year=2018,2019"

response3 = requests.get(undp_url3)

if response3.status_code == 200:
    data3 = response3.json()

df3 = pd.DataFrame(data3)
```

[45]: *# UNDP API Key and Request - BJV*

```
undp_url4 = "https://hdrdata.org/api/CompositeIndices/query?
↳apikey=HDR-K0gLmhxaLjacLvGRi4oTPARBLx54ubNe&year=2016,2017"

response4 = requests.get(undp_url4)

if response4.status_code == 200:
    data4 = response4.json()

df4 = pd.DataFrame(data4)
```

[46]: *# UNDP API Key and Request - BJV*

```
undp_url5 = "https://hdrdata.org/api/CompositeIndices/query?
↳apikey=HDR-K0gLmhxaLjacLvGRi4oTPARBLx54ubNe&year=2014,2015"

response5 = requests.get(undp_url5)

if response5.status_code == 200:
    data5 = response5.json()

df5 = pd.DataFrame(data5)
```

[47]: *# UNDP API Key and Request - BJV*

```
undp_url6 = "https://hdrdata.org/api/CompositeIndices/query?
↳apikey=HDR-K0gLmhxaLjacLvGRi4oTPARBLx54ubNe&year=2012,2013"

response6 = requests.get(undp_url6)

if response6.status_code == 200:
    data6 = response6.json()

df6 = pd.DataFrame(data6)
```

[48]: *# UNDP API Key and Request - BJV*

```
undp_url7 = "https://hdrdata.org/api/CompositeIndices/query?
↳apikey=HDR-K0gLmhxaLjacLvGRi4oTPARBLx54ubNe&year=2010,2011"

response7 = requests.get(undp_url7)

if response7.status_code == 200:
    data7 = response7.json()

df7 = pd.DataFrame(data7)
```

[49]: *# UNDP API Key and Request - BJV*

```
undp_url8 = "https://hdrdata.org/api/CompositeIndices/query?
↳apikey=HDR-K0gLmhxaLjacLvGRi4oTPARBLx54ubNe&year=2008,2009"

response8 = requests.get(undp_url8)

if response8.status_code == 200:
    data8 = response8.json()

df8 = pd.DataFrame(data8)
```

[50]: *# UNDP API Key and Request - BJV*

```
undp_url9 = "https://hdrdata.org/api/CompositeIndices/query?
↳apikey=HDR-K0gLmhxaLjacLvGRi4oTPARBLx54ubNe&year=2006,2007"

response9 = requests.get(undp_url9)

if response9.status_code == 200:
    data9 = response9.json()

df9 = pd.DataFrame(data9)
```

[51]: *# UNDP API Key and Request - BJV*

```
undp_url10 = "https://hdrdata.org/api/CompositeIndices/query?
↳apikey=HDR-K0gLmhxaLjacLvGRi4oTPARBLx54ubNe&year=2004,2005"

response10 = requests.get(undp_url10)

if response10.status_code == 200:
    data10 = response10.json()

df10 = pd.DataFrame(data10)
```


[52]: *# UNDP API Key and Request - BJV*

```
undp_url11 = "https://hdrdata.org/api/CompositeIndices/query?
↳apikey=HDR-K0gLmhxaLjacLvGRi4oTPARBLx54ubNe&year=2002,2003"

response11 = requests.get(undp_url11)

if response11.status_code == 200:
    data11 = response11.json()

df11 = pd.DataFrame(data11)
```

[53]: *# UNDP API Key and Request - BJV*

```
undp_url12 = "https://hdrdata.org/api/CompositeIndices/query?
↳apikey=HDR-K0gLmhxaLjacLvGRi4oTPARBLx54ubNe&year=2000,2001"

response12 = requests.get(undp_url12)

if response12.status_code == 200:
    data12 = response12.json()

df12 = pd.DataFrame(data12)
```

[54]: *# UNDP API Key and Request - BJV*

```
undp_url13 = "https://hdrdata.org/api/CompositeIndices/query?
↳apikey=HDR-K0gLmhxaLjacLvGRi4oTPARBLx54ubNe&year=1998,1999"

response13 = requests.get(undp_url13)

if response13.status_code == 200:
    data13 = response13.json()

df13 = pd.DataFrame(data13)
```

[55]: *# UNDP API Key and Request - BJV*

```
undp_url14 = "https://hdrdata.org/api/CompositeIndices/query?
↳apikey=HDR-K0gLmhxaLjacLvGRi4oTPARBLx54ubNe&year=1996,1997"

response14 = requests.get(undp_url14)

if response14.status_code == 200:
    data14 = response14.json()

df14 = pd.DataFrame(data14)
```

```
[56]: # UNDP API Key and Request - BJV
```

```
undp_url15 = "https://hdrdata.org/api/CompositeIndices/query?
↳apikey=HDR-K0gLmhxaLjacLvGRi4oTPARBLx54ubNe&year=1995"
```

```
response15 = requests.get(undp_url15)
```

```
if response15.status_code == 200:
    data15 = response15.json()
```

```
df15 = pd.DataFrame(data15)
```

```
[57]: all_dataframes = [df1, df2, df3, df4, df5, df6, df7, df8,
                        df9, df10, df11, df12, df13, df14, df15]
```

```
[58]: # Create a list of all data frames to be concatenated. -BJV
```

```
list_of_dfs = [df1, df2, df3, df4, df5, df6, df7, df8,
               df9, df10, df11, df12, df13, df14, df15]
```

```
# Concatenate all the data frames in the list. - BJV
```

```
combined_df = pd.concat(list_of_dfs, ignore_index=True)
```

```
# You can now work with the single combined DataFrame. -BJV
```

```
print(combined_df.head())
```

```
print(f"The final combined data has {len(combined_df)} rows.")
```

```

                country          dimension \
0  AFG - Afghanistan
1  AFG - Afghanistan
2  AFG - Afghanistan  Standard of Living
3  AFG - Afghanistan          Health
4  AFG - Afghanistan

```

```

                                index \
0                                GII - Gender Inequality Index
1                                GII - Gender Inequality Index
2                                MPI - Multidimensional Poverty Index
3                                MPI - Multidimensional Poverty Index
4  PHDI - Planetary pressures-adjusted Human Deve...

```

```

                                indicator  year  value
0  abr - Adolescent Birth Rate (births per 1,000 ...  2022  65.339
1  abr - Adolescent Birth Rate (births per 1,000 ...  2023  64.068
2                                assets - Assets (%)  2022   5.934
3  child_mortality - Child mortality (%)  2022   3.628
4  co2_prod - Carbon dioxide emissions per capita...  2022   0.257

```

```
The final combined data has 178619 rows.
```

```

[59]: # Create a df from a dictionary for saving file - BJV
undp_full_df = pd.DataFrame(combined_df)

[60]: # Country needed to split the ISO3 and the Country names - BJV
split_data = undp_full_df['country'].str.split(' - ', n=1, expand=True)

undp_full_df['ISO3 Code'] = split_data[0]
undp_full_df['Country Name'] = split_data[1]

# Drop the original 'Country' column - BJV
undp_full_df_split = undp_full_df.drop(columns=['country'])

[61]: # 'Value' was causing an error and needed to be converted to numeric - BJV
undp_full_df_split['value'] = pd.to_numeric(undp_full_df_split['value'],
↳ errors='coerce')

# Data needed to be pivoted for clarity when doing EDA - BJV

undp_full_df_pivoted = undp_full_df_split.pivot_table(index=['ISO3 Code',
↳ 'year'],

                                                    columns='indicator',
                                                    values='value')

undp_full_df_pivoted = undp_full_df_pivoted.rename(columns={'ISO3 Code':
↳ 'ISO3'})

print(undp_full_df_pivoted.columns)

```

```

Index(['abr - Adolescent Birth Rate (births per 1,000 women ages 15-19)',
      'assets - Assets (%)', 'child_mortality - Child mortality (%)',
      'co2_prod - Carbon dioxide emissions per capita (production) (tonnes)',
      'coef_ineq - Coefficient of human inequality',
      'cooking_fuel - Cooking fuel (%)',
      'diff_hdi_phdi - Difference from HDI value (%)',
      'drinking_water - Drinking water (%)', 'electricity - Electricity (%)',
      'eys - Expected Years of Schooling (years)',
      'eys_f - Expected Years of Schooling, female (years)',
      'eys_m - Expected Years of Schooling, male (years)',
      'gdi - Gender Development Index (value)', 'gdi_group - GDI Group',
      'gii - Gender Inequality Index (value)', 'gii_rank - GII Rank',
      'gni_pc_f - Gross National Income Per Capita, female (2021 PPP$)',
      'gni_pc_m - Gross National Income Per Capita, male (2021 PPP$)',
      'gnipc - Gross National Income Per Capita (2021 PPP$)',
      'hdi - Human Development Index (value)', 'hdi_f - HDI female',
      'hdi_m - HDI male', 'hdi_rank - HDI Rank', 'housing - Housing (%)',
      'ihdi - Inequality-adjusted Human Development Index (value)',
      'ineq_edu - Inequality in education', 'ineq_inc - Inequality in income',
      'ineq_le - Inequality in life expectancy',

```

```

'le - Life Expectancy at Birth (years)',
'le_f - Life Expectancy at Birth, female (years)',
'le_m - Life Expectancy at Birth, male (years)',
'lfpr_f - Labour force participation rate, female (% ages 15 and older)',
'lfpr_m - Labour force participation rate, male (% ages 15 and older)',
'loss - Overall loss (%)',
'mf - Material footprint per capita (tonnes)',
'mmr - Maternal Mortality Ratio (deaths per 100,000 live births)',
'mpi_value - MPI Value (Range: 0 to 1)',
'mys - Mean Years of Schooling (years)',
'mys_f - Mean Years of Schooling, female (years)',
'mys_m - Mean Years of Schooling, male (years)',
'nutrition - Nutrition (%)',
'phdi - Planetary pressures-adjusted Human Development Index (value)',
'pop_total - Population, total (millions)',
'pr_f - Share of seats in parliament, female (% held by women)',
'pr_m - Share of seats in parliament, male (% held by men)',
'rankdiff_hdi_phdi - Difference from HDI rank',
'sanitation - Sanitation (%)',
'school_attendance - School attendance (%)',
'se_f - Population with at least some secondary education, female (% ages
25 and older)',
'se_m - Population with at least some secondary education, male (% ages
25 and older)',
'years_of_schooling - Years of schooling (%)',
dtype='object', name='indicator')

```

```

[62]: # These are the columns that are relevant to our study from the total column
↳ options - BJV

```

```

columns_to_keep = [
    'hdi - Human Development Index (value)',
    'gii_rank - GII Rank',
    'rankdiff_hdi_phdi - Difference from HDI rank',
    'ihdi - Inequality-adjusted Human Development Index (value)',
    'coef_ineq - Coefficient of human inequality',
    'ineq_le - Inequality in life expectancy',
    'le - Life Expectancy at Birth (years)',
    'ineq_edu - Inequality in education',
    'ineq_inc - Inequality in income',
    'coef_ineq - Coefficient of human inequality'
]

undp_full_df_pivoted.index.rename(
    ['IS03', 'year'],
    inplace=True
)

```

```
undp_df_filtered = undp_full_df_pivoted[columns_to_keep]

undp_df_filtered = undp_df_filtered.reset_index()
```

```
[63]: print("Data retrieval complete: Displaying first 5 rows:")
print(undp_df_filtered.head())

print("\nDataFrame Info:")
undp_df_filtered.info()

print(f"\nDataFrame shape: {undp_df_filtered.shape} (rows, columns)")
```

Data retrieval complete: Displaying first 5 rows:

indicator	ISO3	year	hdi - Human Development Index (value)	\
0	AFG	1995	0.329	
1	AFG	1996	0.334	
2	AFG	1997	0.338	
3	AFG	1998	0.338	
4	AFG	1999	0.347	

indicator	gii_rank - GII Rank	rankdiff_hdi_phdi - Difference from HDI rank	\
0	NaN	NaN	
1	NaN	NaN	
2	NaN	NaN	
3	NaN	NaN	
4	NaN	NaN	

indicator	ihdi - Inequality-adjusted Human Development Index (value)	\
0	NaN	
1	NaN	
2	NaN	
3	NaN	
4	NaN	

indicator	coef_ineq - Coefficient of human inequality	\
0	NaN	
1	NaN	
2	NaN	
3	NaN	
4	NaN	

indicator	ineq_le - Inequality in life expectancy	\
0	NaN	
1	NaN	
2	NaN	
3	NaN	
4	NaN	

```

indicator  le - Life Expectancy at Birth (years)  \
0          52.103
1          52.830
2          53.212
3          52.487
4          54.532

```

```

indicator  ineq_edu - Inequality in education  ineq_inc - Inequality in income  \
0          NaN                                NaN
1          NaN                                NaN
2          NaN                                NaN
3          NaN                                NaN
4          NaN                                NaN

```

```

indicator  coef_ineq - Coefficient of human inequality
0          NaN
1          NaN
2          NaN
3          NaN
4          NaN

```

DataFrame Info:

```
<class 'pandas.core.frame.DataFrame'>
```

RangeIndex: 5974 entries, 0 to 5973

Data columns (total 12 columns):

#	Column	Non-Null Count
0	IS03	5974 non-null
1	year	5974 non-null
2	hdi - Human Development Index (value)	5552 non-null
3	gii_rank - GII Rank	172 non-null
4	rankdiff_hdi_phdi - Difference from HDI rank	156 non-null
5	ihdi - Inequality-adjusted Human Development Index (value)	2297 non-null
6	coef_ineq - Coefficient of human inequality	2304 non-null
7	ineq_le - Inequality in life expectancy	2884 non-null
8	le - Life Expectancy at Birth (years)	5974 non-null

```

    9   ineq_edu - Inequality in education                2543 non-null
float64
   10   ineq_inc - Inequality in income                  2489 non-null
float64
   11   coef_ineq - Coefficient of human inequality      2304 non-null
float64
dtypes: float64(10), object(2)
memory usage: 560.2+ KB

```

DataFrame shape: (5974, 12) (rows, columns)

```
[64]: undp_df_filtered['year'] = pd.to_numeric(undp_df_filtered['year'],
      ↪errors='coerce').astype('Int64')
```

```
[65]: # Joined Work Bank and UNDP Data Frames - BJV

undp_wb_filtered_full = pd.merge(undp_df_filtered, wb_df, on=['IS03', 'year'],
      ↪how='left')
```

```
[66]: # Use pycountry to map country names to ISO3 codes -BJV
iso3_to_name_map = {}
for country in pycountry.countries:
    iso3_to_name_map[country.alpha_3] = country.name

mapped_full_df = pd.DataFrame(undp_wb_filtered_full)

# Add the 'Country Name' column - BJV
mapped_full_df['Country Name'] = mapped_full_df['IS03'].map(iso3_to_name_map)
```

```
[67]: cols = mapped_full_df.columns.tolist()
print(cols)
```

```

['IS03', 'year', 'hdi - Human Development Index (value)', 'gii_rank - GII Rank',
'rankdiff_hdi_phdi - Difference from HDI rank', 'ihdi - Inequality-adjusted
Human Development Index (value)', 'coef_ineq - Coefficient of human inequality',
'ineq_le - Inequality in life expectancy', 'le - Life Expectancy at Birth
(years)', 'ineq_edu - Inequality in education', 'ineq_inc - Inequality in
income', 'coef_ineq - Coefficient of human inequality', 'AG.PR.D.FOOD.XD',
'CC.EST', 'EG.ELC.ACCS.ZS', 'EN.POP.DNST', 'EN.POP.SLUM.UR.ZS',
'ER.H2O.FWST.ZS', 'FX.OWN.TOTL.ZS', 'GB.XPD.RSDV.GD.ZS', 'GE.EST',
'IS.SHP.GCNW.XQ', 'IT.NET.BBND.P2', 'IT.NET.USER.ZS', 'NY.GDP.PCAP.KD.ZG',
'NY.GNP.PCAP.KD.ZG', 'PV.EST', 'RL.EST', 'SE.ADT.LITR.ZS', 'SE.COM.DURS',
'SE.LPV.PRIM', 'SE.LPV.PRIM.SD', 'SE.SEC.CUAT.LO.ZS', 'SE.SEC.CUAT.PO.ZS',
'SE.SEC.CUAT.UP.ZS', 'SE.TER.CUAT.BA.ZS', 'SE.TER.CUAT.DO.ZS',
'SE.TER.CUAT.MS.ZS', 'SE.TER.CUAT.ST.ZS', 'SE.XPD.TOTL.GD.ZS', 'SH.STA.WASH.P5',
'SH.UHC.SRVS.CV.XD', 'SH.XPD.CHEX.PC.CD', 'SH.XPD.GHED.PC.CD', 'SI.POV.GINI',
'SM.POP.TOTL.ZS', 'SN.ITK.MSFI.ZS', 'SP.DYN.LE00.IN', 'SP.RUR.TOTL.ZG',
'VA.EST', 'VC.IHR.PSRC.P5', 'Country Name']

```

```
[68]: current_index_of_country_name = cols.index('Country Name')
```

```
# Pop country name - BJV
```

```
column_to_move = cols.pop(current_index_of_country_name)
```

```
# Move 'country name' to column 2
```

```
cols.insert(1, column_to_move)
```

```
mapped_full_df = mapped_full_df[cols]
```

```
[69]: #save this data frame as an xlsx - BJV
```

```
mapped_full_df.to_excel("undp_wb_mappedfiltered_full2.xlsx", index=True)
```

```
[ ]:
```